

Sample AI + Networking Notes PDF

HTTP Basics

HTTP stands for HyperText Transfer Protocol. It is used for communication between browser and server. When you open a website, the browser sends an HTTP request and the server sends an HTTP response.

HTTPS

HTTPS is the secure version of HTTP. It uses SSL/TLS encryption to protect data between client and server.

Frontend vs Backend

Frontend is what users see on screen. Backend handles APIs, databases, authentication, and business logic.

What is an API?

API stands for Application Programming Interface. APIs allow applications to communicate with each other.

Node.js

Node.js allows JavaScript to run outside the browser. It is commonly used for backend development and APIs.

Express.js

Express is a lightweight backend framework for Node.js. It helps developers create routes, APIs, and middleware easily.

RAG Systems

RAG stands for Retrieval Augmented Generation. In RAG, the system retrieves relevant chunks from documents before sending context to the AI model.

Embeddings

Embeddings are numerical vector representations of text. They help AI systems understand semantic meaning instead of exact keywords.

Vector Databases

Vector databases store embeddings. They help perform similarity search between user questions and document chunks.

Concept	Meaning
HTTP	Browser-server communication protocol
RAG	Retrieve relevant context before AI answer
Embedding	Meaning represented as vectors
Chunking	Splitting large text into smaller pieces